MPI Programming Lab.

(MPI → Message Passing Interface)

1) Install WSL

2) sudo apt update↵

3) sudo apt install : openmpi-bin libopenmpi-dev↵

4) Version check

mpicc --version

mpirun --version

First_mpi.c

```
#include <stdio.h>
#include <mpi.h>

int main(int argc, char *argv[])
{
    int rank, size;

    MPI_Init(&argc, &argv);

    MPI_Comm_rank(MPI_COMM
    _WORLD, &rank);

    MPI_Comm_size(MPI_COMM
    _WORLD, &size);
    printf("Hello from process %d out
```

```
of %d processes \n", rank, size);
MPI_finalize();
return 0;
}
```

Explanation

size → It assigns no. of processes

rank → It assigns process id of each process

What is MPI?

→ It means message passing interface. In these proging, each process communicates with each other by assigning small datas in private memory of each process.

What is Open MP?

→ It works as shared memory.

Run command

```
mpicc filename.c -o filename ↵
mpirun -np 4 ./filename
```

↑ No. of processes

Open MP Programming

(Open Multi - Processing)

It is a shared - memory parallel programming model based on compiler directives and designed for multi-core CPU.

In memory-wise comparison, MPI has distributed memory but open-MP has shared memory.

In open-MP, the process runs depending on compiler directives instead of explicit message passing.

On the basis of difficulty level, MPI is hard but openMP is quit-easy.

Applications :-

Best - fit :

Open MP execution model :-

① It is a fork-join model. It starts with a thread, called master-thread.

② Threads are forked.

③ Threads work in parallel.

④ Threads join back into one.

Why synchronization is needed in open MP?

→ In openmp, threads share memory. So they may read or write the same variable. This can cause wrong result, race condition and inconsistent output. To ensure correctness, sync. is needed.